

04.11.15(E)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam

Nov- Dec - 2015

Max. Marks: 100

Class : SE

Name of the Course: **Discrete Structures and Graph Theory.**

Course Code: UCEC305

Duration: 3Hours

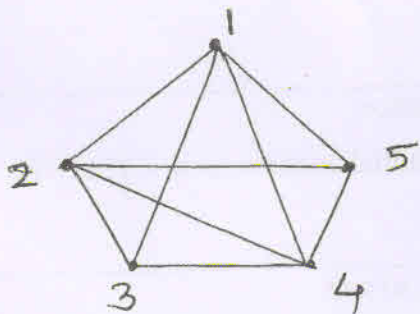
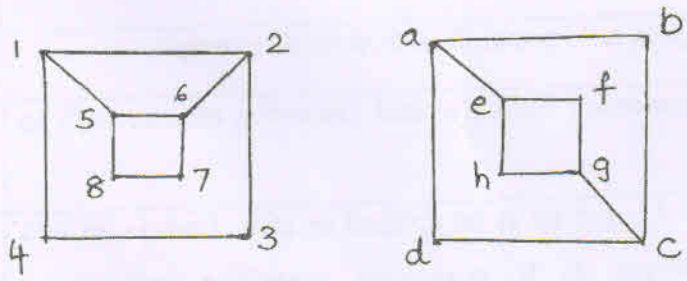
Semester: III

Branch: COMP

Instructions:

1. All Questions are Compulsory

Question No.		Max. Marks
Q 1	a) Determine the number of integers between 1 and 250 that are divisible by any of the integers 2, 3, 5 and 7.	5
	b) If $A \rightarrow B$ is one to one and onto, then f inverse is both one to one and onto. Prove.	5
	c) Prove that $5^n - 1$ is divisible by 4, for $n \geq 1$ by mathematical induction.	5
	d) Solve the recurrence relation $a_n = 6a_{n-1} - 9a_{n-2}$ with initial conditions $a_0 = 1, a_1 = 6$.	5
Q 2	a) Let $A = (a, b, c, d, e)$ and $R = ((a,a), (a,b), (a,c), (b,c), (c,e), (c,d), (d,e), (a,d))$ Compute R^2, R^3 and R^4 Draw diagrams of R^2, R^3 and R^4 .	10
	b) Explain pigeon hole principle with suitable example.	5
	c) Define Symmetric, Reflexive and Transitive closure with an example.	5
Q 3	a) Let $A = Z$, and let R be defined as $aRb \mid a-b$ is an odd positive integer. Is R symmetric, transitive, reflexive, antisymmetric, an equivalence relation. Justify.	10

	b) Let $f(x) = 2x - 3$, $g(x) = x^2 + 3x + 5$. Find $g \circ f$ and $f \circ g$.	05
	c) Let $A = \{1, 2, 3, 4\}$ and relation $R = \{(1, 1), (1, 2), (1, 4), (2, 4), (3, 1), (3, 2), (4, 2), (4, 3), (4, 4)\}$ Find the transitive closure of R by Warshall's Algorithm.	05
Or		
Q 3	a) Let $A = (1, 2, 3, 4)$, R is a relation defined as aRb if $b \leq a+1$. Determine whether R is reflexive, irreflexive, symmetric, asymmetric, antisymmetric?	5
	b) Find the complement of each element in D_{42} .	5
	c) Explain universal and existential quantifiers, tautology with examples	5
	d) Construct the Hasse diagram of D_{30} . Is it a Lattice? If yes, check the distributive property.	5
Q 4	Determine whether the given graph is planar. Justify with Euler formula.	5
		
	a) Determine whether the following graphs are isomorphic?	5
		

	b) A connected planer graph has 9 vertices having degrees 2, 2, 2, 3, 3, 3, 4, 4, 5. How many edges are there ?	5
	c) Define Hamiltonian path and circuit. Find Hamiltonian path and circuit.	5
Q 5	a) Let $H = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ Be a parity check matrix, Determine the group code $e_H : B^3 \rightarrow B^6$	10
	b) Consider the (2,5) group encoding function $e : B^2 \rightarrow B^5$ defined by $e(00) = 00000$ $e(01) = 01110$ $e(10) = 10101$ $e(11) = 11011$ Decode the following words relative to maximum likelihood decoding function. i) 11110 ii) 10011 iii) 10100	05
	b) Show that a group G is Abelian if and only if $(ab)^2 = a^2b^2$ for all elements a and b in G .	05

Or		
Q 5	a) Consider the (3, 7) encoding function $e : B^3 \rightarrow B^7$ defined by $e(000) = 0000000$ $e(100) = 1000101$ $e(001) = 0010110$ $e(101) = 1010011$ $e(010) = 0101000$ $e(110) = 1101101$ $e(011) = 0111110$ $e(111) = 1111011$ is a group code How many errors it can detect?	10
	b) Define group, monoid, semigroup. Determine whether the set together with the binary operation $*$ is a semigroup, monoid or a group ? Set of real numbers with $a*b = a + b + 2$. Set of $m * n$ matrices under the operation of multiplication	10